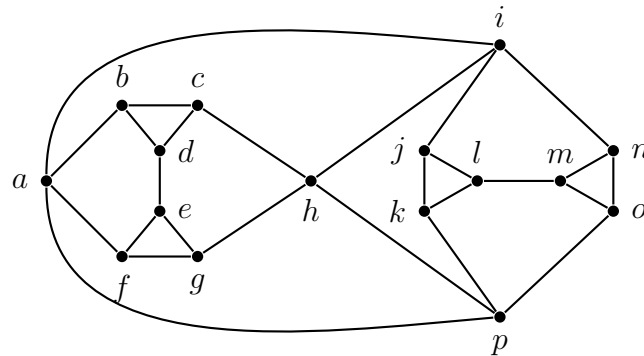
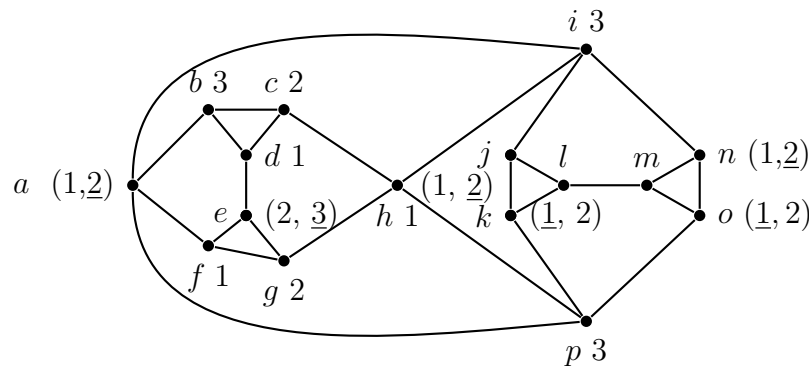


2.3

1.(r) Find the chromatic number of the following graph

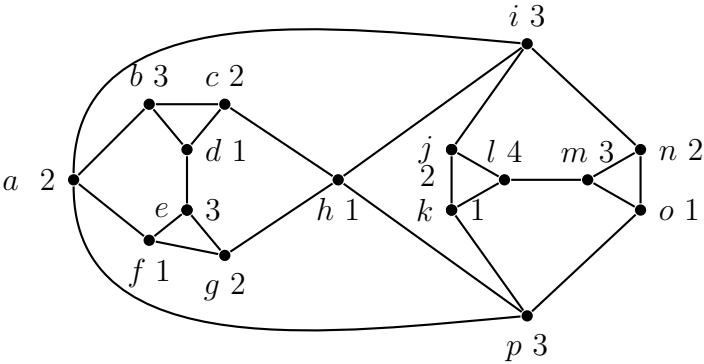


* $\chi(G) \geq 3$ b/c a K_3 is contained in the graph



- WLOG color d, c, b by 1, 2, 3
- **Choose** $e : 3, a : 2$
- **Force** $f : 1, g : 2$
- **Choose** $h : 1$
- **Force** $i, p : 3$
- **Choose** $n, j : 2$ and $o, k : 1$
- Cannot be done because l is adjacent to m and both are adjacent to a 1 and 2, l or m : 4
- **Backtrack** to $h : 1$, **Force** $h : 3$
- Cannot be done because l or m are still forced to be 4
- **Backtrack** to $e : 3, a : 2$, **Force** $a : 1, e : 2$
- Cannot be done because l or m are still forced to be 4

$$\Rightarrow \chi(G) \geq 4$$



$\Rightarrow \boxed{\chi(G) = 4}$